Pedestrian attribute recognition under domain shift

UPAR 2027 Track 1

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From recognition to transfer

We chose UPAR 2027 Track 1, which asks models to predict 40 binary pedestrian attributes from person crops, including age, gender, hair, clothing and accessories. We wanted to know how well a strong classifier trained on familiar datasets would handle unfamiliar surveillance images. Starting with a semantic SigLIP2 baseline, we tested domain sampling, added DINOv3 features, tried limited adaptation and examined a separate robustness augmentation policy.

Only the original model has a confirmed official result: **scored 0.770450 Challenge Avg.** The later ensembles improved on every locally held-out domain, but a competition-site outage prevented their planned official evaluation. Their local results suggest better transfer; their official performance remains unknown.

Why we used leave-one-domain-out evaluation

At a threshold of 0.5, the original model scored 0.856185 on known-domain validation, compared with 0.770450 in the official evaluation. That gap led us to focus on transfer. We introduced three leave-one-domain-out (LODO) folds, training on two datasets and evaluating on the third. Within each fold, source-training, source-selection and source-calibration partitions were disjoint.

Each fold started from fresh general-purpose foundation weights. Label frequencies, sampling, checkpoint selection and calibration used only source data, and we fixed the checkpoint and thresholds before evaluating the target. The six-epoch reference folds differed substantially in size: 18,668 training images when PA100K was held out, 89,001 for PETA and 87,669 for Market1501. Six epochs therefore meant different numbers of updates and did not ensure comparable convergence.

Once we had inspected the LODO targets, later comparisons became development experiments. They remain controlled comparisons, but they are no longer untouched tests and do not establish statistical certainty. We distinguish official, LODO and source-selection scores throughout.

Two complementary visual representations

SigLIP2: semantic queries over pedestrian patches

The baseline uses general-purpose SigLIP2-Base-NaFlex. Aspect-aware preprocessing produces at most 256 valid 16×16 patches. We initialize 40 attribute queries from pretrained text embeddings, giving labels such as clothing color and accessories a semantic starting point. The text encoder is used only for initialization.

An eight-head cross-attention layer lets each query attend to the valid visual patches. Query residuals, a feed-forward network and normalization produce an attribute-specific representation. Each attribute has its own linear classifier for local logits. A second linear branch uses the masked spatial mean to add global evidence; padded patches are excluded from both attention and pooling.

$$z_a = w_a^T h_a + b_a + [W_g \, mean_{valid}(X)]_a$$

X denotes visual patch features, and h_a is the residual cross-attention representation for attribute a. Dropout is applied during training.

DINOv3: appearance features with regional attention

We chose the self-supervised DINOv3 ViT-B/16 to complement SigLIP2's language-aligned features. Direct checkpoint access was unavailable, but a maintainer-supported public timm release of the original pretrained model passed our provenance and license checks.

Component	DINOv3 implementation
Input	Aspect-preserving 320×160 canvas; ImageNet normalization
Token layout	1 CLS + 4 register tokens + 200 spatial patches (20×10)
Six descriptors	CLS; spatial mean; head mean; upper-body mean; lower-body mean; spatial maximum
Overlapping regions	Head 0-30%; upper 15-65%; lower 50-100% of resized person height
Padding and registers	Means weighted by patch/content overlap; registers never spatially pooled
Attribute head	LayerNorm → learned attention over six descriptors → 40 independent logits

$$\alpha_{ak} = softmax_k(q_a^T d_k/\sqrt{768});\ z_a = w_a^T(\Sigma_{k=1\ldots6}\alpha_{ak} d_k) + b_a$$

d_k is a normalized descriptor. Each attribute learns which regions to use. The overlapping bands avoid the need for a pose estimator, and the classifier uses dropout 0.2 during training.

Candidate A freezes the entire DINO backbone and trains only the head on cached FP16 descriptors. Candidate B starts from the original foundation weights and the appropriate frozen head, then updates Transformer blocks 10/11, final normalization and the head. Every LODO head is trained strictly on source data. The all-domain head is reserved for final all-domain training.

Shared learning and scoring rules

Masked, class-balanced binary cross-entropy

For image i and attribute a, y_{ia} is a binary label, v_{ia} marks a finite valid 0/1 label, and z_{ia} is the logit. We compute prevalence p_a from valid source-training labels before sampling, then use it to set exponential weights for positive and negative labels.

$$p_a = \frac{\Sigma_i v_{ia} y_{ia}}{\Sigma_i v_{ia}};\ w_a^+ = exp(1 - p_a);\ w_a^- = exp(p_a)$$

Rare positives receive a larger positive weight. Sampling probabilities do not change the frequencies used to calculate these weights.

$$L = \frac{1}{max(1,V)} \Sigma_{i,a} v_{ia}[w_a^+ y_{ia}\ softplus(-z_{ia}) + w_a^-(1 - y_{ia})\ softplus(z_{ia})]$$

$V = \Sigma_{i,a}\ v_{ia}$ and softplus(t) = log(1 + exp(t)). The denominator counts valid labels, not weighted labels. Invalid targets are sanitized before elementwise BCE is evaluated.

Domain sampling and probability blending

$$weight(image\ i\ in\ domain\ d) \propto N_d^{\alpha-1} \Rightarrow P(domain\ d) \propto N_d^{\alpha}$$

The original α = 0.5 gives square-root-domain sampling. Setting α = 0 in the ablation gives each source domain equal probability. Images are sampled with replacement.

$$p_{ensemble,a} = 0.75\,\sigma(z_{SigLIP,a}) + 0.25\,\sigma(z_{DINO,a});\ \hat{y}_a = 1[p_{ensemble,a} > t_a]$$

The fixed mixture combines raw sigmoid probabilities, before binarization or separate calibration. We fit 40 global thresholds using source-calibration data only.

Recipe	Trainable parameters / schedule	Learning rates and batch
Original SigLIP2	Head warmup; upper 4 blocks + norm; 12 epochs; EMA 0.9995; best epoch 7	AdamW: backbone 1e-5, head 1e-4; effective batch 64
SigLIP LODO / C	Same architecture and loss; 6 epochs per fold; C changes augmentation only	Original optimizer and sampler; seed 2027
Frozen DINO / A	Head only; max 30 epochs; patience 5; LODO best 30/27/25	AdamW head 3e-4; batch 128; dropout 0.2
Adapted DINO / B	Blocks 10/11 + norm + head; 6 LODO epochs; no EMA introduced	AdamW backbone 5e-6, head 5e-5; batch 16×4 = 64

How we calculate the score

$$mA = mean_a[\frac{TPR_a + TNR_a}{2}];\ F1 = \frac{2PR}{P + R};\ C_d = \frac{2\,mA_d F1_d}{mA_d + F1_d}$$

P and R are mean-image precision and recall over valid attributes. Primary F1 is their harmonic mean, rather than the mean of per-image F1 scores. Local mA excludes attributes without both positive and negative support. Macro is the mean of the three domain scores; worst is their minimum.

Calibration uses two coordinate passes over 11 fixed thresholds from 0.22 to 0.78, with a 0.005 mean absolute deviation penalty from 0.5. The organizer's exact empty-label conventions are private. We considered sensitivities to alternative F1 definitions and unsupported labels, but cannot claim complete equivalence between our local scorer and the official one.

What the controlled comparisons measured

Calibrated method	PA100K	PETA	Market	Macro	Worst
SigLIP2 baseline	0.797104	0.777228	0.817448	0.797260	0.777228
Equal-domain	0.797565	0.776778	0.813926	0.796089	0.776778
Frozen DINO alone	0.754206	0.724861	0.783061	0.754043	0.724861
A: frozen ensemble	0.802375	0.780285	0.820231	0.800964	0.780285
B: adapted DINO alone	0.780643	0.752457	0.817344	0.783481	0.752457
B: adapted ensemble	0.803448	0.781356	0.822720	0.802508	0.781356
C: robustness	0.797197	0.776418	0.820613	0.798076	0.776418

All values are local Challenge Avg under the common three-fold development protocol. Each target was excluded from its fold's training, selection and calibration. Six decimal places identify the recorded results; they do not imply statistical certainty.

Calibration helped, but transfer remained uneven

Source-only calibration raised the baseline's LODO macro from 0.789605 to 0.797260, with gains on all three targets. PETA remained the weakest domain at 0.777228. Calibration transferred usefully, though visual domain shift remained a problem. A disjoint known-source calibration audit scored 0.863482, close to the in-sample calibrated value of 0.864952. That small amount of optimism could not explain the much larger gap to the official result.

Every fold had both positive and negative source examples for all 40 attributes. Some labels were still scarce: with PA100K held out, lower-orange had only four source positives. Age, accessories and lower clothing colors remained difficult. We could not explain these errors simply as missing source labels, or assume that all of them came from the backbone.

Equal-domain sampling did not help

We suspected that the large PA100K source might dominate learning. To test this, we changed only α from 0.5 to 0, making the two source domains equally likely while keeping class-frequency calculations and BCE weights unchanged. We logged actual sampling draws for the new run. The baseline's draws had not been logged, so its exposure probabilities are expected values.

Calibrated macro fell by 0.001171 (0.117 percentage points), and worst-domain performance fell by 0.000451. PA100K improved slightly, but PETA regressed and Market1501 dropped by 0.003522. Raw macro also fell, from 0.789605 to 0.786694, so calibration was not hiding a consistent gain.

Neither predeclared adoption margin, +0.003 for macro and +0.003 for the worst domain, was met. We kept square-root sampling. Repeating a smaller source more often may have reduced exposure to the larger source's diversity, but a single-seed experiment cannot establish that explanation.

The standalone DINO results tell us how well it works as a replacement for SigLIP; the ensemble results tell us whether it adds useful information. We also distinguish an average gain accompanied by a weaker worst domain from improvement across all domains. These comparisons use the same folds and scoring conventions.

Complementarity helped more than replacement

Calibrated macro	mA	Instance-F1	Precision	Recall
SigLIP2 baseline	0.785674	0.809226	0.787004	0.833028
A: frozen ensemble	0.786367	0.816127	0.794574	0.839026
B: adapted ensemble	0.788772	0.816744	0.794937	0.839878

Candidate A: useful in a fixed ensemble

Frozen DINO scored only 0.754043 macro on its own. Its errors differed enough from SigLIP's to make it useful in combination: probability-residual correlations were approximately 0.747-0.776. The predeclared 75/25 blend improved all three domains. Against SigLIP, macro rose by 0.003704 (0.370 percentage points), and worst-domain performance rose by 0.003057.

Most of the gain came from instance-F1 (+0.006901), with a smaller increase in mA (+0.000693). Precision and recall both improved. PETA's balanced accuracy nevertheless declined despite its higher Challenge Avg. Age and accessories generally benefited, while clothing-color results were mixed. The models complemented each other without improving every attribute or group.

A's macro of 0.800964 fell short of the predeclared 0.8023 promotion threshold. Its worst domain exceeded 0.7772 and no domain regressed, but the promotion criterion was still unmet. The complementary signal was promising enough to justify one controlled partial-adaptation experiment. We did not search for new ensemble weights.

Candidate B: limited adaptation improved all three domains

We unfroze the final two DINO blocks, final normalization and the head. Each fold used its corresponding source-trained frozen head, with original pretrained weights elsewhere. All folds completed six epochs, and source selection chose epochs 6/5/5. Standalone macro increased to 0.783481, still below SigLIP, which kept the case for using DINO in an ensemble.

Compared with A, the B ensemble gained 0.001073 on PA100K, 0.001070 on PETA and 0.002489 on Market1501. Macro reached 0.802508, and worst-domain performance reached 0.781356. Macro mA rose by 0.002405 and instance-F1 by 0.000617. Adaptation therefore contributed more through balanced accuracy than the earlier frozen blend had.

B's macro gain over A was 0.001544, below its predeclared +0.002 advancement margin. Applying A's older absolute threshold would have changed the criterion after the result was known. We recorded B as failing its research advancement criterion. The later all-domain B run was authorized separately as an exploratory competition candidate.

What we selected

We kept the original SigLIP2 checkpoint as the scored fallback and retained the fixed 75/25 mixture for A and B. B was our strongest local transfer candidate. We used no hidden-test scores or target-based searches for thresholds or mixture weights to choose these settings. The gains across all three domains were encouraging, although they came from one seed and targets we had inspected repeatedly.

Final candidates, setbacks and lessons

Candidate C: robustness augmentation was rejected

This independent SigLIP branch changed only augmentation. The bounded policy used down/up sampling (p=0.30, scale 0.65-1), JPEG (p=0.30, quality 60-95), blur (p=0.15, σ 0.2-0.7), gamma (p=0.20, 0.9-1.1) and brightness (p=0.50, 0.9-1.1). We intended to approximate surveillance degradation while preserving geometry and avoiding strong color distortion.

C reached 0.798076 macro, only 0.000816 above baseline, while PETA, the weakest domain, fell to 0.776418. Neither +0.003 advancement margin was met. We rejected C and did not train an all-domain version. Mild degradation may have removed evidence for small accessories or fine color distinctions, but we did not demonstrate that this caused the regression.

Final all-domain A and B

A's frozen head trained on all 97,669 training images, with epoch 30 selected. Final B started from the original DINO foundation and A's all-domain head, then adapted the same components as in LODO. New ensemble thresholds were fitted only on calibration labels. The original production SigLIP weights and thresholds stayed separate and unchanged.

Final B epoch	Raw DINO source-selection score	Change from preceding row
6	0.838134	-
7	0.839221	+0.001087
8 (selected)	0.840005	+0.000783

Training finished at eight epochs, and epoch 8 was selected. The source gains were still positive but shrinking. They justify investigating limited continuation, though they do not show that more epochs would improve unseen-domain performance. We ran no further extension.

Evaluation context	A ensemble	B ensemble	Interpretation
Final source selection	0.867806	0.869305	Known-source development; not LODO
Calibrated LODO macro	0.800964	0.802508	Transfer comparison on three held-out domains
Official evaluation	Not confirmed	Not confirmed	Only original SigLIP2: 0.770450

The original submission, **returned official scores of 0.773647 mA, 0.767279 instance-F1, 0.741799 precision and 0.807915 recall, alongside 0.770450 Challenge Avg**. The A/B results above are local measurements from different evaluation settings.

The follow-up upload for A reset, and reconnection failed during the reported competition-site outage. We could not complete the remaining planned official evaluations. A and B had improved on every locally held-out domain; B gained 0.005248 macro and 0.004127 on the worst domain over baseline. We therefore expect better generalization, but only an official evaluation could test that expectation. Adding these local gains to 0.770450 would not give a valid official score.

Challenges, reproducibility and next steps

The repository includes working code, compact settings and aggregate results; private images, features and weights are excluded. Across our experiments, combining complementary representations helped more reliably than equalizing domain exposure or adding generic degradation. The next step is to repeat the fixed comparisons across seeds and match training updates across source datasets of different sizes. Rare-attribute coverage and source-only calibration transfer also need further work. Limited continuation of B is worth testing while monitoring mA and worst-domain performance. The current evidence does not establish a universally better model or a higher official score for A or B.